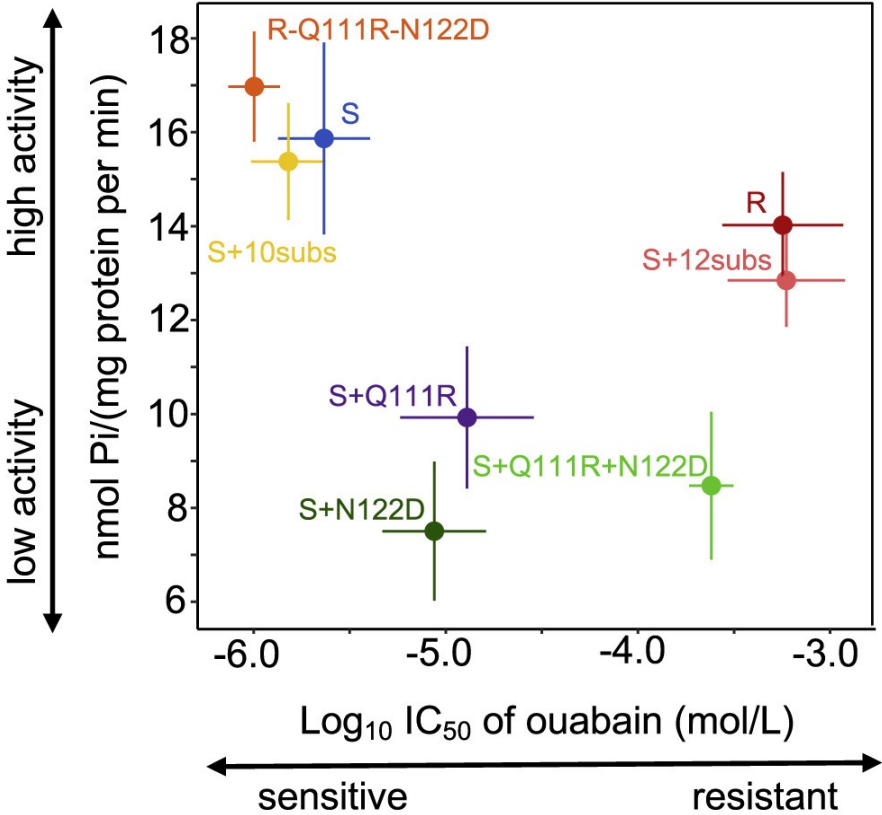


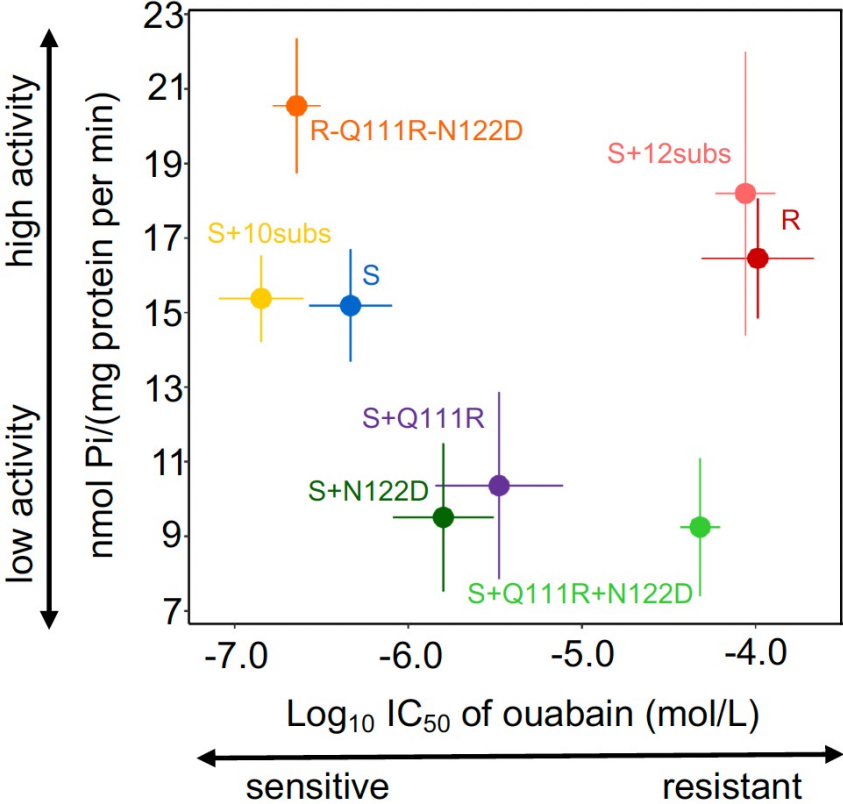
Original Results, Fig 5B

B



Corrected Results, Fig 5B

B



Original text, Fig 5B

We then quantified the level of CTS resistance of each genotype using enzyme-inhibition assays ([Table S4](#); [Figure S6](#)).³⁶ Individually, Q111R and N122D significantly increased CTS resistance by 21-fold and 14-fold, respectively (ANOVA $p = 2.7\text{e-}13$ and $p = 2.3\text{e-}6$; [Figure 5B](#); [Table S5](#)). When combined, Q111R and N122D produce a greater than 100-fold increase in CTS resistance relative to the S paralog (Tukey's HSD test; adjusted $p < 4\text{e-}5$; [Figure 5B](#); [Tables S5](#) and [S6](#)). In contrast, the remaining 10 substitutions had no detectable net effect on CTS resistance when jointly added to the S background ($p = 0.22$; [Figure 5B](#)).

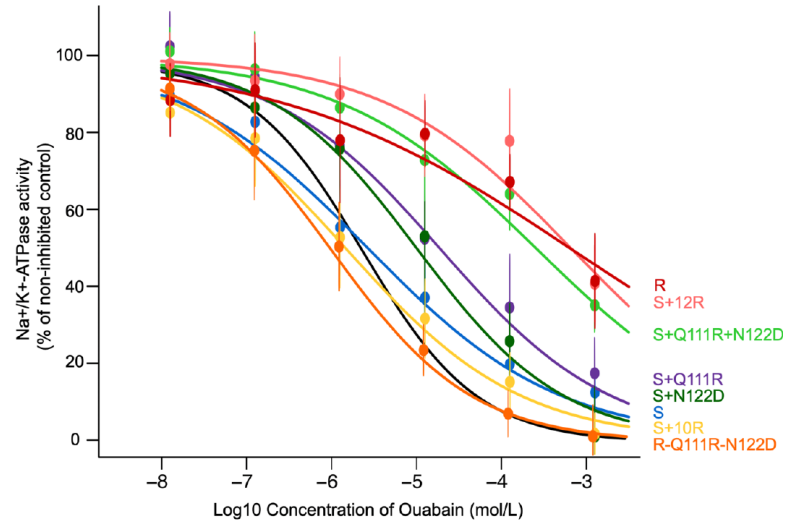
Given the absence of detectable effects of R/S-distinguishing substitutions other than Q111R and N122D on CTS resistance, we tested whether these substitutions had effects on other aspects of ATP1A1 function. Because ATP hydrolysis and ion co-transport are strongly coupled functions of Na⁺,K⁺-ATPase,³⁷ we used estimates of the rate of ATP hydrolysis in the absence of ouabain as a proxy for overall protein activity. Based on this assay, we found that CTS resistance substitutions Q111R and N122D significantly impair activity, individually reducing ATPase activity by an average of 40% ($p = 0.024$ and $p = 7.7\text{e-}4$, respectively; [Figure 5B](#); [Table S5](#)). We also detected a significant interaction between Q111R and N122D that renders their joint effects somewhat less severe than predicted by the sum of their individual effects (i.e., a 30% reduction rather than the expected 78% reduction; $p = 0.022$). Critically, adding the remaining 10 R-specific substitutions on the S background containing Q111R and N122D restores ATPase activity close to S levels—a significant effect even when controlling for the effects of Q111R and N122D (ANOVA $p = 1\text{e-}4$; [Figure 5B](#); [Table S5](#)). Our results thus indicate that these 10 R/S-distinguishing substitutions play a vital role in compensating for the negative pleiotropic effects of the resistance-conferring substitutions, Q111R and N122D. We conclude that the evolution of the R protein from a CTS-sensitive ancestral state involved two epistatically interacting substitutions (Q111R and N122D) in conjunction with compensatory effects of 10 additional substitutions that mitigate the trade-off between toxin resistance and native enzyme activity.

Corrected text, Fig 5B

We then quantified the level of CTS resistance of each genotype using enzyme-inhibition assays ([Table S4](#); [Figure S6](#)).³⁶ Individually, Q111R and N122D significantly increased CTS resistance by 7-fold and 3.5-fold, respectively (ANOVA $p = 2\text{e-}13$ and $p = 6\text{e-}6$; [Figure 5B](#); [Table S5](#)). When combined, Q111R and N122D produce a greater than 100-fold increase in CTS resistance relative to the S paralog (Tukey's HSD test; adjusted $p < 4\text{e-}5$; [Figure 5B](#); [Tables S5](#) and [S6](#)). In contrast, the remaining 10 substitutions had no detectable net effect on CTS resistance when jointly added to the S background (ANOVA $p = 0.61$; [Figure 5B](#)).

Given the absence of detectable effects of R/S-distinguishing substitutions other than Q111R and N122D on CTS resistance, we tested whether these substitutions had effects on other aspects of ATP1A1 function. Because ATP hydrolysis and ion co-transport are strongly coupled functions of Na⁺,K⁺-ATPase,³⁷ we used estimates of the rate of ATP hydrolysis in the absence of ouabain as a proxy for overall protein activity. Based on this assay, we found that CTS resistance substitutions Q111R and N122D substantially impair activity, individually reducing ATPase activity by an average of 35% ([Figure 5B](#); [Table S5](#)) and S proteins bearing Q111R, N122D or both have significantly lower activity relative to all other constructs ($p = 4.8\text{e-}5$, nested ANOVA). Notably, the joint effects of Q111R and N122D are less severe than predicted by the sum of their individual effects (i.e., a 39% reduction rather than the expected 69% reduction). Critically, adding the remaining 10 R-specific substitutions on the S background containing Q111R and N122D restores ATPase activity close to S levels—a significant effect even when controlling for the effects of Q111R and N122D (ANOVA $p = 2.4\text{e-}4$; [Figure 5B](#); [Table S5](#)). Our results thus indicate that these 10 R/S-distinguishing substitutions play a vital role in compensating for the negative pleiotropic effects of the resistance-conferring substitutions, Q111R and N122D. We conclude that the evolution of the R protein from a CTS-sensitive ancestral state involved two epistatically interacting substitutions (Q111R and N122D) in conjunction with compensatory effects of 10 additional substitutions that mitigate the trade-off between toxin resistance and native enzyme activity.

Original, Fig S6



Corrected, Fig S6

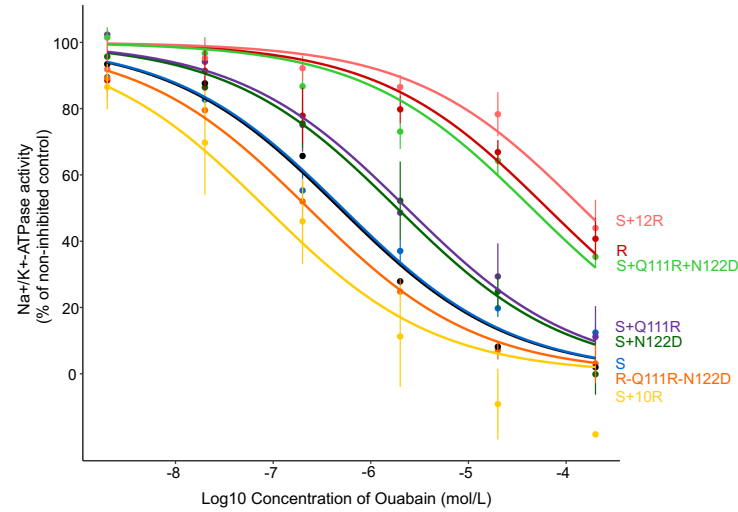


Figure S6. Cardiotonic steroid (ouabain) inhibition curves for six each engineered

Leptodactylus Na⁺,K⁺-ATPase produced in this study, related to Figure 5. Points and error bars represent the mean $\pm$ SEM (n=6 biological replicates) percentage of protein activity relative to controls measured in the absence of ouabain and excluding the activity of background ATPases. The black inhibition curve was measured from commercially procured porcine cerebral cortex (CAS 9000-83-3, Sigma-Aldrich, Inc.) and represents a standard benchmark reference for cardiotonic steroid sensitivity (Log₁₀ IC₅₀ = -5.61). The ATP1B1 of *Leptodactylus macrosternum* was co-expressed with each engineered version of ATP1A1 (Table S5).

Original, Table S4

Construct Name	Engineered Substitution(s)	Description	Ouabain sensitivity (mol/L) Mean(log10 IC ₅₀) ± SD	ATPase activity nmol Pi/(mg protein*min) ± SD
S	-	Sensitive (S) paralog of <i>L. macrosternum</i> ATP1A1	-5.63 ± 0.59	16.30 ± 5.01
S+Q111R	Q111R	Q111R on the S paralog background	-4.89 ± 0.85	10.68 ± 3.71
S+N122D	N122D	N122D on the S paralog background	-5.06 ± 0.66	7.16 ± 3.62
S+Q111R+N122D	Q111R + N122D	Q111R and N122D on the S paralog background	-3.62 ± 0.28	8.29 ± 3.86
S+10subs	A112T, E116D, I135V, L180Q, I199L, I279V, S403C, L536M, Q701L, I788M	All substitutions strongly distinguishing R and S paralogs, except Q111R and N122D, on the S paralog background	-5.82 ± 0.47	16.37 ± 3.05
R-Q111R-N122D	R111Q, D122N	Reversions R111Q and D122N on the R paralog background	-5.60 ± 0.33	17.41 ± 2.87
S+12subs	Q111R, A112T, E116D, N122D, I135V, L180Q, I199L, I279V, S403C, L536M, Q701L, I788M	Twelve substitutions strongly distinguishing R and S paralogs on the S paralog background	-3.23 ± 0.75	12.17 ± 2.43
R	-	Resistant (R) paralog of <i>L. macrosternum</i> ATP1A1	-3.25 ± 0.77	14.09 ± 2.77

Table S4. List of engineered ATP1A1 constructs used to test functional effects of amino acid substitutions in *Leptodactylus* including summary of the ouabain sensitivity and catalytic properties of Na⁺,K⁺-ATPase for each ATP1A1 construct, related to Figure 5. The values represent the mean and standard deviation (SD) ouabain sensitivity (log₁₀IC₅₀) of ATPase activity of six biological replicates. ATP1B1 of *Leptodactylus macrosternum* was co-expressed with ATP1A1.

Note: R and S paralogs of *L. macrosternum* differ by the 12 substitutions that are the focus of this study and by 9 additional amino-acid substitutions and a two-amino acid insertion-deletion difference. Our experiments revealed that these 10 *L. macrosternum*-specific substitutions do not contribute detectably to S vs. R differences in CG resistance of enzyme function (using all 10 as one co-variate, ANOVA p>0.5. Following convention, positions of substitutions are standardized relative to the sheep (*Ovis aries*) sequence NM_001009360 - 5 AA from 5' end.

Corrected, Table S4

Construct Name	Engineered Substitution(s)	Description	Ouabain sensitivity (mol/L) Mean(log10 IC ₅₀) ± SD	ATPase activity nmol Pi/(mg protein*min) ± SD
S	-	Sensitive (S) paralog of <i>L. macrosternum</i> ATP1A1	-6.33 ± 0.58	15.19 ± 3.69
S+Q111R	Q111R	Q111R on the S paralog background	-5.48 ± 0.90	10.35 ± 6.15
S+N122D	N122D	N122D on the S paralog background	-5.80 ± 0.71	9.51 ± 4.87
S+Q111R+N122D	Q111R + N122D	Q111R and N122D on the S paralog background	-4.32 ± 0.28	9.25 ± 4.53
S+10subs	A112T, E116D, I135V, L180Q, I199L, I279V, S403C, L536M, Q701L, I788M	All substitutions strongly distinguishing R and S paralogs, except Q111R and N122D, on the S paralog background	-6.85 ± 0.60	15.38 ± 2.83
R-Q111R-N122D	R111Q, D122N	Reversions R111Q and D122N on the R paralog background	-6.64 ± 0.34	20.54 ± 4.43
S+12subs	Q111R, A112T, E116D, N122D, I135V, L180Q, I199L, I279V, S403C, L536M, Q701L, I788M	Twelve substitutions strongly distinguishing R and S paralogs on the S paralog background	-4.06 ± 0.39	18.04 ± 10.42
R	-	Resistant (R) paralog of <i>L. macrosternum</i> ATP1A1	-4.00 ± 0.79	16.45 ± 3.96

Original, Table S5

(Explanatory Variables) ANOVA	Ouabain sensitivity				ATPase activity			
	log10(IC ₅₀)				nmol Pi/(mg protein*min)			
	df	MS	F	p value	df	MS	F	p value
Q111R	1, 42	42.8	107.8	2.7e-13	1, 42	83.0	6.9	0.015
N122D	1, 42	11.8	27.72	2.3e-6	1, 42	101.4	7.98	7.2e-3
10subs	1, 42	0.59	1.6	0.22	1, 42	228.1	17.96	1.2e-4
R-S background	1, 42	0.04	1.9	0.74	1, 42	11.5	0.34	0.34
Q111R:N122D	-	-	-	-	1, 42	7.6	5.64	0.022

(Explanatory Variables) Linear regression	Ouabain sensitivity				ATPase activity			
	log10(IC ₅₀)				nmol Pi/(mg protein*min)			
	Est	SE	t	p value	Est	SE	t	p value
Intercept	-6.03	0.17	-36.3	<2e-16	14.3	1.19	12.1	3e-15
Q111R	1.32	0.21	6.27	2.7e-13	-4.39	1.88	-2.34	0.024
N122D	1.14	0.21	5.45	2.3e-6	-6.81	1.88	-3.63	7.7e-4
10subs	0.26	0.22	1.19	0.24	1.94	1.46	17.96	0.18
R-S background	-0.08	0.26	-0.34	0.74	1.39	1.46	0.34	0.35
Q111R:N122D	-	-	-	-	6.90	2.91	5.64	0.022

Table S5. Statistical analysis of ouabain sensitivity and ATPase activity, related to Figure 5. Significant p values are highlighted in bold.

Note: “R-S background” in the ANOVA refers to 9 additional amino acid substitutions and a two amino acid insertion-deletion difference that distinguishes the R and S constructs (derived from *Leptodactylus macrosternum*).

Corrected, Table S5

(Explanatory Variables) ANOVA	Ouabain sensitivity				ATPase activity			
	log10(IC ₅₀)				nmol Pi/(mg protein*min)			
	df	MS	F	p value	df	MS	F	p value
Q111R	1	43.6	111.8	2.1e-13	1	39.23	1.3	0.26
N122D	1	10.4	10.4.5	6.2e-6	1	28.79	1.0	0.34
10 subs	1	0.1	0.27	0.61	1	488.8	16.1	2.4e-4
R-S background	1	0.1	0.34	0.56	1	23.7	0.8	0.38
Residuals	42	16.4			42	30.4		

Table S5. Statistical analysis of ouabain sensitivity and ATPase activity, related to Figure 5. Significant p values are highlighted in bold. Interaction terms were not significant.

Note: “10 subs” refers to the 10 background substitutions identified as being targets of selection. “R-S background” refers to the remaining 9 additional amino acid substitutions and a two amino acid insertion-deletion difference that distinguishes the R and S constructs (derived from *Leptodactylus macrosternum*).